

Remote Photoplethysmography (rPPG) Deep Learning Pipeline

Comprehensive Read-Only Diagnostic & Model Failure Analysis Report

Target Model: rPPGCNN (3D-CNN)	Evaluated Checkpoint: best_rppg_model.pth (300.42 MB)
Dataset: UBFC-rPPG (DATASET_2)	Temporal Window: 150 Frames (5.0s @ 30 FPS)
Validation Samples: 71 Windows (6 Subjects)	Diagnostic Status: 100% Read-Only / Non-Invasive

Executive Summary & Core Finding

Observed Evaluation Metric: Validation MSE = **0.996567** | Prediction Range = **[-0.1126, +0.1044]** | Target Range = **[-2.1299, +3.8337]**.

Root Cause Summary: The model in best_rppg_model.pth collapsed to a trivial zero-mean predictor because of a **total Dying ReLU failure** in the dense bottleneck layer (relu_fc). Exactly **100% (256/256)** of the intermediate neurons output zero, completely severing backpropagation gradients to all 3D convolution layers. The model output is mathematically identical to the static bias vector of the final layer: $y_pred = b_fc2$.

1. Dataset Quality & Integrity Audit

All preprocessed NumPy arrays in checkpoints/ were loaded and verified for numerical integrity. Data tensors are clean with zero NaNs, zero Infs, and valid physiological bounds.

Array Name	Tensor Shape	Dtype	Min	Max	Mean	Std Dev	NaN / Inf
X_train	(362, 3, 150, 64, 64)	float32	0.0000	0.9961	0.4851	0.1983	0 / 0
y_train_norm	(362, 150)	float32	-2.4673	4.3894	-1.21e-8	1.0000	0 / 0
X_val	(71, 3, 150, 64, 64)	float32	0.0000	0.9961	0.4746	0.1873	0 / 0
y_val	(71, 150)	float32	-2.1299	3.8337	-1.14e-8	1.0000	0 / 0

2. Physiological Target & Video Input Dynamics

Target BVP Quality: Every ground-truth BVP window exhibits non-zero peak-to-peak pulse morphology. Training peak-to-peak range spans **3.05 to 6.10** (mean 3.86); Validation spans **3.08 to 5.19** (mean 3.88). Signals are active, healthy photoplethysmography waveforms standardized to mean ≈ 0 , std = 1.0.

Input Video Dynamics: Face ROI pixel intensities change smoothly across frames with average consecutive frame difference of **0.0051** and temporal pixel standard deviation of **0.0190** (min 0.0058, max 0.0915). No static, frozen, or corrupted windows exist in the dataset.

3. Subject Independence & Split Validation

The subject split strictly enforces zero leakage across sets: **29 Training subjects** (362 windows, ~ 12.5 windows/subject), **6 Validation subjects** (71 windows, ~ 11.8 windows/subject: subject1, 4, 9, 10, 35, 39), and **7 Test subjects**. Subject overlap between training and validation is strictly **ZERO**.

4. Backpropagation Gradient Diagnostic

A diagnostic single-pass backward propagation was performed on a sample batch without updating weights. The gradient norms across every parameter layer reveal a complete gradient freeze:

Layer Parameter	Tensor Shape	Parameter Count	Gradient L2 Norm	Status
conv1.weight / bias	[16, 3, 3, 3, 3] / [16]	1,312	0.000000	FROZEN (Zero Grad)
conv2.weight / bias	[32, 16, 3, 3, 3] / [32]	13,856	0.000000	FROZEN (Zero Grad)
conv3.weight / bias	[64, 32, 3, 3, 3] / [64]	55,360	0.000000	FROZEN (Zero Grad)
fc1.weight / bias	[256, 307200] / [256]	78,643,456	0.000000	FROZEN (Zero Grad)
fc2.weight	[150, 256]	38,400	0.000000	FROZEN (Zero Grad)
fc2.bias	[150]	150	0.097773	ACTIVE (Updating)

5. Empirical Proof of Dying ReLU Failure

Tracing activations through the feed-forward pass reveals the exact layer where signal transmission collapses:

Processing Stage	Output Tensor Shape	Mean Activation	Active Units (> 0)	State Description
conv1 + pool1	(4, 16, 150, 32, 32)	0.115802	100%	Healthy feature maps
conv2 + pool2	(4, 32, 150, 16, 16)	0.060435	100%	Healthy feature maps
conv3 + pool3	(4, 64, 75, 8, 8)	0.001654	100%	Low-magnitude activations
Flattened Input	(4, 307,200)	0.001654	100%	307,200 inputs to FC1
fc1 Pre-Activation	(4, 256)	-0.054422	0.00% (100% <= 0)	All pre-activations negative
fc1 Post-ReLU	(4, 256)	0.000000	0 / 1024 (0.00%)	100% DEAD RELU
fc2 Output (Prediction)	(4, 150)	-0.001288	N/A	Output = b_fc2 only

6. Architecture Breakdown & Capacity Imbalance

The network architecture exhibits extreme structural imbalance between convolutional and dense layers:

Layer Subsystem	Component Details	Parameter Count	% of Model
3D Convolutional Feature Extractor	Conv3D (16, 32, 64 filters) + MaxPool3D	70,528	0.09 %
Dense Flatten Layer (fc1)	Linear(307,200 -> 256) + ReLU + Dropout(0.5)	78,643,456	99.86 %
Output Linear Projection (fc2)	Linear(256 -> 150)	38,550	0.05 %
TOTAL MODEL CAPACITY	Full rPPGCNN Parameter Count	78,752,534	100.00 %

Capacity-to-Data Ratio: With 78,752,534 parameters trained on only 362 training samples, the model operates at 217,548 parameters per single training sample. Over 99.8% of the weights reside in a single unregularized fully-connected matrix connecting 307,200 flattened features.

7. Notebook Training History Analysis

The training trajectory from rPPG_training.ipynb (Cell 35) reveals that the model stalled at Epoch 1 and made virtually zero optimization progress over 50 full epochs:

Training Epoch	Training Loss (MSE)	Validation Loss (MSE)	Observed Network Behavior
Epoch 01	1.009556	1.000979	Initial loss $\approx \text{Var}(Y) = 1.0$; dying ReLU occurs instantly
Epoch 10	1.000330	0.999461	All weights frozen; only output bias shifts slightly
Epoch 25	0.998520	0.997857	Plateau continues; zero gradient backpropagation
Epoch 50 (Final)	0.997020	0.996567	Saved as best_rppg_model.pth (Total $\Delta\text{Val} = 0.0044$)

8. Ranked Failure Diagnosis & Mechanistic Causes

Rank 1: Total Dying ReLU in Dense Bottleneck (Primary Mechanistic Failure)

When 307,200 flattened features are fed into fc1 (256 neurons), the cumulative weighted sum across 307,200 dimensions pushes all 256 neuron pre-activations into negative values (mean = -0.0544, min = -0.8953, 100% ≤ 0). Standard ReLU maps all negative inputs to exactly 0.0 with a derivative of 0.0. This completely severed backpropagation, leaving all 3D convolutions with zero gradient.

Rank 2: Giant Flatten Dimension Without Spatio-Temporal Pooling

Modern video rPPG architectures (e.g. PhysNet, DeepPhys, TS-CAN) use **Adaptive Spatial/Temporal Average Pooling** or 1x1 convolutions before projection to collapse (C, T, H, W) to (C, T, 1, 1). Directly flattening (64, 75, 8, 8) generated 307,200 unstructured features, destroying spatio-temporal coherence and exploding parameter count.

Rank 3: Extreme Overparameterization & Capacity Mismatch

78.75 million parameters cannot be trained on 362 video windows without severe optimization pathologies. Over 99.8% of the weights reside in the single fc1.weight matrix, dominating the loss landscape and preventing convolutional feature discovery.

Rank 4: Trivial MSE Global Minimum Convergence

For any zero-mean, unit-variance standardized target Y , the expected error of predicting a constant zero is $E[(Y - 0)^2] = \text{Var}(Y) = 1.000$. When backpropagation fails, the optimizer minimizes MSE simply by adjusting the output bias vector to the global mean ($b_{fc2} \approx 0$), achieving an MSE of ~ 0.9965 without learning any pulse representation.

Report Status: Audit Complete | **Inspection Type:** Non-Invasive Read-Only Verification | **Artifact Generated:** rPPG_Pipeline_Diagnostic_Report.pdf